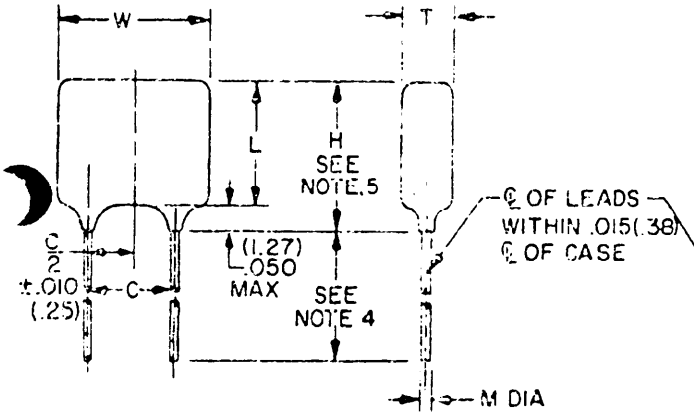
MIL-C-11015/24E
8 April 1974
SUPERSEDING
MIL-C-11015/24D
14 November 1972

MILITARY SPECIFICATION SHEET

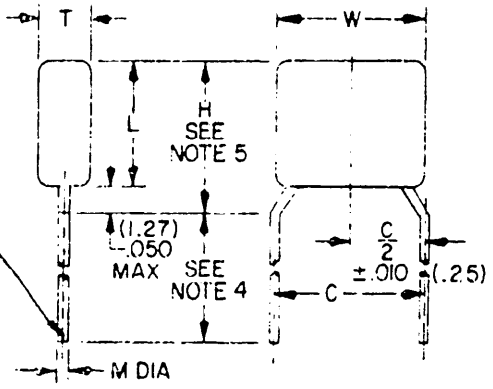
CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLES CK41, CK42, CK43, CK50, CK51, CK52, CK53, CK54 AND CK55

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.



STYLES CK42 AND CK43



STYLE CK41

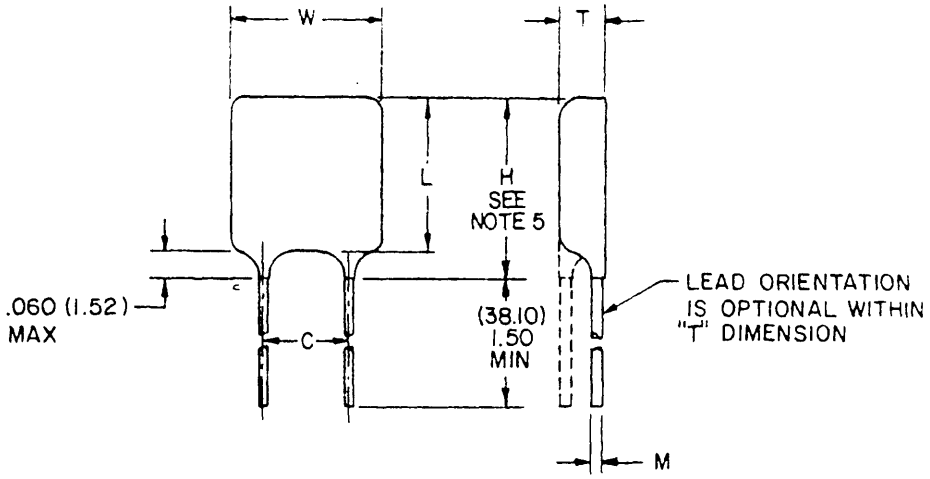
INACTIVE FOR NEW DESIGN
STYLES CK41, CK42 and CK43

Dimensions						
Style	W ±.020 (.51)	L ±.020 (.51)	T ±.015 (.38)	C ±.010 (.25)	M +.004 (.10) -.001 (.03)	H (Ref)
CK41	.080 (2.03)	.090 (2.29)	.045 (1.14)	.100 (2.54)	.016 (.41)	.160 (4.06)
CK42	.130 (3.30)	.140 (3.56)	.045 (1.14)	.100 (2.54)	.016 (.41)	.210 (5.33)
CK43	.230 (5.84)	230 (5.84)	.045 (1.14)	.200 (5.08)	.016 (.41)	.300 (7.62)

FIGURE 1. Styles CK41, CK42 and CK43.

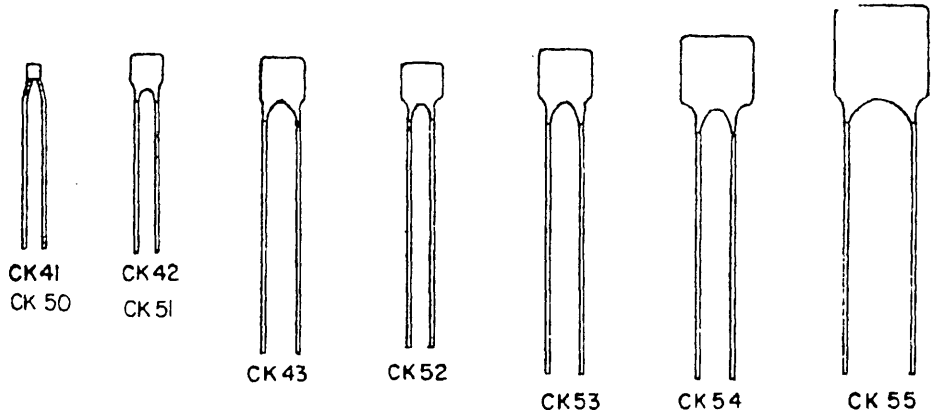
FSC 5910

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Dimensions						
Style	W ± .020 (.51)	L ± .020 (.51)	T +.025 (.64) -.050 (1.27)	C ± .020 (.51)	M +.006 (.15) -.001 (.03)	H (Ref)
~~CK50~~	~~.080 (2.03)~~	.080 (2.03)	.075 (1.91)	.100 (2.54)	.016 (.41)	.170 (4.32)
CK51	.130 (3.30)	.130 (3.30)	.075 (1.91)	.100 (2.54)	.016 (.41)	.200 (5.08)
CK52	.180 (4.57)	.180 (4.57)	.100 (2.54)	.100 (2.54)	.020 (.51)	.250 (6.35)
CK53	.280 (7.11)	.280 (7.11)	.125 (3.18)	.200 (5.08)	.020 (.51)	.350 (8.89)
CK54	.380 (9.65)	.380 (9.65)	.125 (3.18)	.200 (5.08)	.020 (.51)	.450 (11.43)
CK55	.480 (12.19)	.480 (12.19)	.175 (4.45)	.400 (10.16)	.025 (.64)	.550 (13.97)

FIGURE 2. Styles CK50, CK51, CK52, CK53, CK54 and CK55.



ACTUAL SIZE (APPROXIMATE)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Lead length shall be .750 (19.05 mm) to 1.50 (38.10 mm) for styles CK41 and CK42 and 1.25 (31.75 mm) to 1.50 (38.10 mm) for style CK43.
5. H dimension is for reference purposes to indicate the maximum overall mounting height of the capacitor. Leads shall be solderable beyond this point.

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TABLE I. Styles CK41, CK42, CK43, CK50, CK51, CK52, CK53, CK54 and CK55.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK41BX1R0D	100	BX	1.0	D
1R5D			1.5	D
2R2D			2.2	D
3R3D			3.3	D
4R7D			4.7	D
5R6K			5.6	K
6R8K			6.8	K
8R2K			8.2	K
100-			10	K, M
120K			12	K
150-			15	K, M
180K			18	K
220-			22	K, M
270K			27	K
330-			33	K, M
390K			39	K
470-			47	K, M
560K			56	K
680-			68	K, M
820K			82	K
101-			100	K, M
121K			120	K
151-			150	K, M
181K			180	K
221-			220	K, M
271K			270	K
331-			330	K, M
391K			390	K
471-			470	K, M
561K			560	K
681-			680	K, M
821K			820	K
CK41BX102-			1,000	K, M
CK42BX122K			1,200	K
CK42BX152-			1,500	K, M
CK42BX182K			1,800	K
CK43BX222-			2,200	K, M
CK43BX272K	100		2,700	K
CK50BX2R2D	200		2.2	D
2R7D			2.7	D
3R3D			3.3	D
3R9D			3.9	D
4R7D			4.7	D
5R6K			5.6	K
6R8K			6.8	K
8R2K			8.2	K
CK50BX100-	200	BX	10	K, M

See footnote at end of table.

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TABLE I. Styles CK41, CK42, CK43, CK50, CK51, CK52, CK53, CK54 and CK55 - Continued.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK50BX120K	200	BX	12	K
150-	↕	↕	15	K, M
180K			18	K
220-			22	K, M
270K			27	K
330-			33	K, M
390K			39	K
470-			47	K, M
560K			56	K
680-			68	K, M
820K			82	K
101-			100	K, M
121K			120	K
151-			150	K, M
181K			180	K
221-			220	K, M
271K			270	K
331-	↓		330	K, M
391K	100		390	K
471-	↕		470	K, M
561K			560	K
681-			680	K, M
821K			820	K
102-			1,000	K, M
122K			1,200	K
152-			1,500	K, M
182K	↓		1,800	K
222-	50		2,200	K, M
272K	↕		2,700	K
332-			3,300	K, M
339K			3,900	K
472-			4,700	K, M
CK50BX562K	50		5,600	K
CK51BX391K	200		390	K
471-	↕		470	K, M
561K			560	K
681-			680	K, M
821K			820	K
102-			1,000	K, M
122K	↓		1,200	K
222-	100		2,200	K, M
272K	↕		2,700	K
332-			3,300	K, M
392K			3,900	K
472-		↓	4,700	K, M
CK51BX562K	100	BX	5,600	K

See footnote at end of table.

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TABLE I. Styles CK41, CK42, CK43, CK50, CK51, CK52, CK53, CK54 and CK55 - Continued.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK51BX682-	50	BX	6,800	K, M
822K	↑	↑	8,200	K
103-	↕	↕	10,000	K, M
123K	↓	↓	12,000	K
CK51BX153-	50		15,000	K, M
CK52BX152-	200		1,500	K, M
182K	↑		1,800	K
222-	↕		2,200	K, M
272K	↕		2,700	K
332-	↕		3,300	K, M
392K	↕		3,900	K
472-	↕		4,700	K, M
562K	↕		5,600	K
682-	↕		6,800	K, M
822K	↓		8,200	K
103-	200		10,000	K, M
123K	100		12,000	K
153-	↑		15,000	K, M
183K	↕		18,000	K
223-	↕		22,000	K, M
273K	↓		27,000	K
333-	100		33,000	K, M
393K	50		39,000	K
473-	↑		47,000	K, M
563K	↕		56,000	K
683-	↓		68,000	K, M
CK52BX823K	50		82,000	K
CK53BX123K	200		12,000	K
153-	↑		15,000	K, M
183K	↕		18,000	K
223-	↕		22,000	K, M
273K	↕		27,000	K
333-	↕		33,000	K, M
393K	↕		39,000	K
473-	200		47,000	K, M
563K	100		56,000	K
683-	↑		68,000	K, M
823K	↕		82,000	K
104-	↕		100,000	K, M
124K	↕		120,000	K
154-	100		150,000	K, M
184K	50		180,000	K
224-	↑		220,000	K, M
274K	↓	↓	270,000	K
CK53BX334-	50	BX	330,000	K, M

See footnote at end of table.

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TABLE I. Styles CK41, CK42, CK43, CK50, CK51, CK52, CK53, CK54 and CK55 - Continued.

Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	volts, dc		pF	
CK54BX563K	200	BX	56,000	K
683-	↕	↕	68,000	K, M
823K	↕	↕	82,000	K
104-	200	↕	100,000	K, M
184K	100	↕	180,000	K
224-	↕	↕	220,000	K, M
274K	↕	↕	270,000	K
334-	↕	↕	330,000	K, M
394K	↕	↕	390,000	K
474-	100	↕	470,000	K, M
564K	50	↕	560,000	K
CK54BX684-	50	↕	680,000	K, M
CK55BX124K	200	↕	120,000	K
154-	↕	↕	150,000	K, M
184K	↕	↕	180,000	K
224-	200	↕	220,000	K, M
564K	100	↕	560,000	K
684-	↕	↕	680,000	K, M
824K	↕	↕	820,000	K
105-	100	↕	1,000,000	K, M
125K	50	↕	1,200,000	K
155-	↕	↕	1,500,000	K, M
185K	↕	↕	1,800,000	K
CK55BX225-	50	BX	2,200,000	K, M

1/ When applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figures 1 and 2.
Case type - Dipped or molded.
Capacitance value - See table I.
Capacitance tolerance - ±.5 pF (D), ±10 percent (K), or ±20 percent (M) as shown in table I.
Rated temperature - -55° C to +125° C.

Dielectric withstanding voltage:
Dielectric (DWV): Method 301 of MIL-STD-202.
Test voltage - 250 percent of rated voltage.
Body insulation: In accordance with MIL-C-11015.
Test potential - 500 volts dc.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition D (100,000 ft).
Test voltage - 150 percent of rated voltage.

Insulation resistance (IR): Method 302 of MIL-STD-202 (measured with rated voltage applied), 100,000 megohms, min, or 1,000 megohm-microfarads, min, whichever is less.

Capacitance (Cap.): Method 305 of MIL-STD-202.
Test frequency:
100 pF or less - 1 MHz ±1 kHz.
Over 100 pF - 1 kHz ±100 Hz.
Test voltage - 1.0 ±0.2 volt rms or referred thereto.

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Dissipation factor (DF): 2.5 percent, max.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

Temperature cycling and immersion:
Temperature cycling: Method 102 of MIL-STD-202, condition D, except step 3 shall be at +125° C.
Immersion: Method 104 of MIL-STD-202, condition B.
DWV - 250 percent of rated voltage.
IR - Not less than 50 percent of initial +25° C requirement.

Salt spray (corrosion): Not applicable.

Terminal strength: Method 211 of MIL-STD-202, condition A.
Applied load - 3-pound radial pull for all styles except CK54 and CK55 which shall be 5 pounds.

Moisture resistance: Method 106 of MIL-STD-202, 20 cycles (rated voltage applied first 10 cycles).
DWV - 250 percent of rated voltage.
IR - Not less than 50 percent of initial +25° C requirement for styles CK41, CK42 and CK43.
Not less than 10 percent of initial +25° C requirement for styles CK50, CK51, CK52, CK53, CK54 and CK55.
Cap. - Within ±10 percent of initial measured value.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
IR - Not less than initial +25° C requirement.
Cap. - Within ±5 percent of initial measured value.
DF - 3.0 percent, max.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature):
Method 108 of MIL-STD-202, 2,000 hours (for qualification); 250 hours (performance check); and 1,750 hours (continuation test).
Test potential - 200 percent of rated voltage.
IR:
At +125° C before life test - 10,000 megohms, min, or 100 megohm-microfarads, min, whichever is less.
At +25° C - Not less than 50 percent of initial +25° C requirement for styles CK41, CK42 and CK43.
Not less than 20 percent of initial +25° C requirement for styles CK50, CK51, CK52, CK53, CK54 and CK55.
At -125° C - Not less than 50 percent of +125° C requirement for styles CK41, CK42 and CK43.
Not less than 20 percent of +125° C requirement for styles CK50, CK51, CK52, CK53, CK54 and CK55.
DF - 2.5 percent, max.

Marking: Styles CK43, CK53, CK54 and CK55 shall be marked in accordance with MIL-C-11015. Styles CK42, CK51, and CK52 shall be marked in accordance with the following:
Face - The two-digits of style and letter symbol of capacitance tolerance.
Example - 52D
Reverse - Three digits designating capacitance value.
Example: 2R2
Styles CK41 and CK50 shall be marked with three color dots (two significant and one multiplier) in accordance with MIL-STD-1285 to specify capacitance in picofarads.

Supersession data:
Styles CK07 and CK08 formerly covered by this specification sheet have been deleted. Recommended substitute items are listed in table II.
Styles CK43 (capacitance values of 3,300 pF and above), CK44 and CK45 have been deleted. Since there has never been a qualified source for these items, no supersession data is offered.

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TABLE II. Substitute items.

Superseded item	Recommended substitute item	Superseded item	Recommended substitute item
CK07BX2R2D	CK41BX2R2D	CK07BX181K	CK41BX181K
CK07BX3R3D	CK41BX3R3D	CK07BX221-	CK41BX221-
CK07BX3R9D	CK41BX3R9D	CK07BX271K	CK41BX271K
CK07BX4R7D	CK41BX4R7D	CK07BX331-	CK41BX331-
CK07BX5R6D	CK41BX5R6D	CK07BX391K	CK41BX391K
CK07BX6R8K	CK41BX6R8K	CK07BX471-	CK41BX471-
CK07BX8R2K	CK41BX8R2K	CK07BX561K	CK41BX561K
CK07BX100-	CK41BX100-	CK07BX681-	CK41BX681-
CK07BX120K	CK41BX120K	CK07BX821K	CK41BX821K
CK07BX150-	CK41BX150-	CK07BX102-	CK41BX102-
CK07BX180K	CK41BX180K	CK08BX122K	CK52BX122K
CK07BX220-	CK41BX220-	CK08BX152-	CK52BX152-
CK07BX270K	CK41BX270K	CK08BX182K	CK52BX182K
CK07BX330-	CK41BX330-	CK08BX222-	CK52BX222-
CK07BX390K	CK41BX390K	CK08BX272K	CK52BX272K
CK07BX470-	CK07BX470-	CK08BX332-	CK52BX332-
CK07BX560K	CK41BX560K	CK08BX392K	CK52BX392K
CK07BX680-	CK41BX680-	CK08BX472-	CK52BX472-
CK07BX820K	CK41BX820K	CK08BX562K	CK52BX562K
CK07BX101-	CK41BX101-	CK08BX682-	CK52BX682-
CK07BX121K	CK41BX121K	CK08BX822K	CK52BX822K
CK07BX151-	CK41BX151-	CK08BX103-	CK52BX103-

Changes from previous issue: Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL, MI
Navy - OS
Air Force - 11, 17, 80, 85
DSA - ES

User activities:
Army -
Navy - AS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-1100)

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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
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